DP014 **DAQ SET A**
Physics 1
Semester 1
Session 2023/2024
2 hours

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PROGRAM DRAW A QUESTION

MODEL PHYSICS QUESTION PAPER

2 HOURS

DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DO SO.

INSTRUCTIONS TO CANDIDATE:

The question paper consists of **7** question.

Answer **all** questions.

The use of non-programmable scientific calculator is permitted.

QUESTION	MARKS
1	
2	
3	
4	
5	
6	
7	
TOTAL	

LIST OF SELECTED CONSTANT VALUES
SENARAI NILAI PEMALAR TERPILIH

Speed of light in vacuum *Laju cahaya dalam vakum*	c	$= 3.00 \times 10^{8}\ \mathrm{m\ s^{-1}}$
Permeability of free space *Ketelapan ruang bebas*	μ_o	$= 4\pi \times 10^{-7}\ \mathrm{H\ m^{-1}}$
Permittivity of free space *Ketelusan ruang bebas*	ε_o	$= 8.85 \times 10^{-12}\ \mathrm{F\ m^{-1}}$
Electron charge magnitude *Magnitud cas elektron*	e	$= 1.60 \times 10^{-19}\ \mathrm{C}$
Planck constant *Pemalar Planck*	h	$= 6.63 \times 10^{-34}\ \mathrm{J\ s}$
Electron mass *Jisim elektron*	m_e	$= 9.11 \times 10^{-31}\ \mathrm{kg}$ $= 5.49 \times 10^{-4}\ \mathrm{u}$
Neutron mass *Jisim neutron*	m_n	$= 1.674 \times 10^{-27}\ \mathrm{kg}$ $= 1.008665\ \mathrm{u}$
Proton mass *Jisim proton*	m_p	$= 1.672 \times 10^{-27}\ \mathrm{kg}$ $= 1.007277\ \mathrm{u}$
Deuteron mass *Jisim deuteron*	m_d	$= 3.34 \times 10^{-27}\ \mathrm{kg}$ $= 2.014102\ \mathrm{u}$
Molar gas constant *Pemalar gas molar*	R	$= 8.31\ \mathrm{J\ K^{-1}\ mol^{-1}}$
Avogadro constant *Pemalar Avogadro*	N_A	$= 6.02 \times 10^{23}\ \mathrm{mol^{-1}}$
Boltzmann constant *Pemalar Boltzmann*	k	$= 1.38 \times 10^{-23}\ \mathrm{J\ K^{-1}}$
Free-fall acceleration *Pecutan jatuh bebas*	g	$= 9.81\ \mathrm{m\ s^{-2}}$
Atomic mass unit *Unit jisim atom*	1 u	$= 1.66 \times 10^{-27}\ \mathrm{kg}$ $= 931.5\ \dfrac{M\ eV}{c^2}$
Electron volt *Elektron volt*	1 eV	$= 1.6 \times 10^{-19}\ \mathrm{J}$

LIST OF SELECTED CONSTANT VALUES
SENARAI NILAI PEMALAR TERPILIH

Constant of proportionality for Coulomb's law *Pemalar hukum Coulomb*	$k = \dfrac{1}{4\pi\varepsilon_o}$	$= 9.0 \times 10^9\ \mathrm{N\,m^2\,C^{-2}}$
Atmospheric pressure *Tekanan atmosfera*	1 atm	$= 1.013 \times 10^5\ \mathrm{Pa}$
Density of water *Ketumpatan air*	ρ_w	$= 1000\ \mathrm{kg\,m^{-3}}$

LIST OF SELECTED FORMULAE
SENARAI RUMUS TERPILIH

1. $v = u + at$
2. $s = ut + \frac{1}{2}at^2$
3. $v^2 = u^2 + 2as$
4. $s = \frac{1}{2}(u + v)t$
5. $v = u - gt$
6. $v^2 = u^2 - 2gs$
7. $s = ut - \frac{1}{2}gt^2$
8. $p = mv$
9. $J = F\Delta t$
10. $J = \Delta p = mv - mu$
11. $f_s \leq \mu_s N$
12. $f_k \leq \mu_k N$
13. $W = \vec{F} \cdot \vec{s} = Fs\cos\theta$
14. $K = \frac{1}{2}mv^2$
15. $U = mgh$
16. $U_s = \frac{1}{2}kx^2 = \frac{1}{2}Fx$
17. $a_c = \frac{v^2}{r} = r\omega^2 = v\omega$
18. $F_c = \frac{mv^2}{r} = mr\omega^2 = mv\omega$
19. $s = r\theta$
20. $v = r\omega$
21. $a_t = r \propto$
22. $\omega = \omega_o - \alpha t$
23. $\omega^2 = {\omega_o}^2 - 2\alpha\theta$
24. $\theta = \omega_o t - \frac{1}{2}\alpha t^2$
25. $\theta = \frac{1}{2}(\omega_o + \omega)t$
26. $\tau = rF\sin\theta$
27. $\frac{Q}{t} = -kA\left(\frac{\Delta T}{L}\right)$
28. $v_{rms} = \sqrt{\langle v^2 \rangle}$
29. $v_{rms} = \sqrt{\frac{3kT}{m}} = \sqrt{\frac{3RT}{M}}$
30. $\Delta U = Q - W$

Answer all the questions

1.

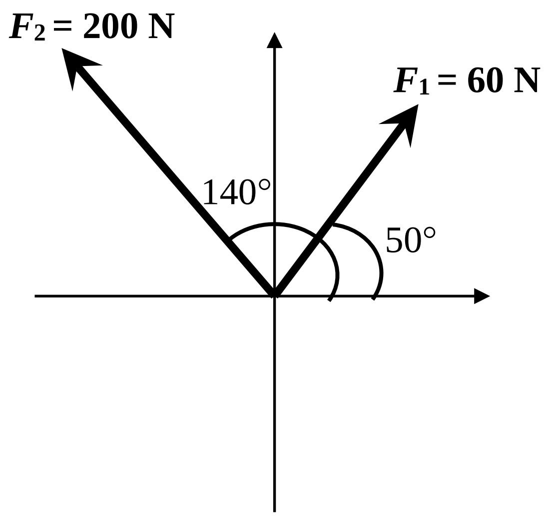

FIGURE 1

FIGURE 1 shows F_1 is 60 N at 50° and F_2 is 200 N at 140°. Determine the magnitude and direction of the resultant vector.

[8 *marks*]

2. (a) A train takes 30 minutes to travel from Padang Tengku to Kerambit station. The train accelerates in 60 s from rest at 0.5 m s^{-2}. Then, it travels at a constant speed in 6 min **and** decelerates with constant deceleration before it stops.

(i) Sketch a velocity – time graph for the journey.

(ii) Determine the speed for 60 s, the deceleration, **and** the distance from Padang Tengku to Kerambit station.

[10 *marks*]

(b) A van is moving with a velocity of 10 m s^{-1}. The driver accelerated at 3 m s^{-2}. Will he be able to get to his destination which is 800 m away within 35 s?

[3 *marks*]

3. (a) A 0.060 kg ball is thrown to the right straight toward a wall with a speed of 10 m s^{-1}. It rebounds straight backward at a speed of 8 m s^{-1}. If the ball is in contact with the wall for 3.0 ms, determine

(i) the impulse on the ball.

(ii) the average force exerted on the ball by the wall.

[4 *marks*]

(b)

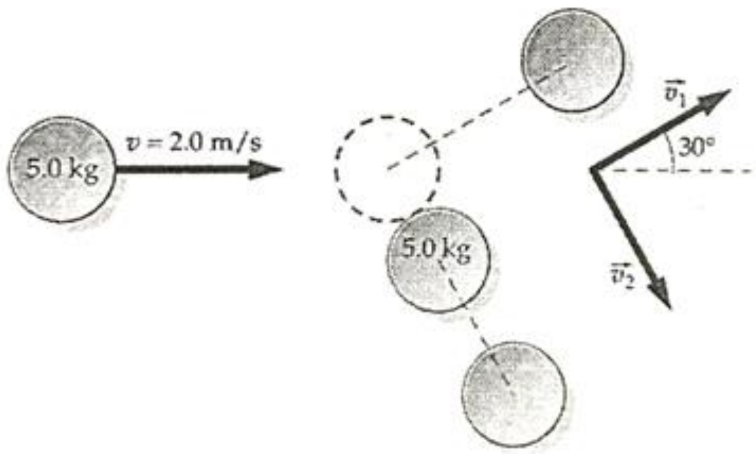

FIGURE 3

FIGURE 3 shows a rubber disc of mass 5.0 kg moving at 2.0 m s^{-1} approaches an identical rubber disc that is stationary on a frictionless surface. After the collision, the first disc leaves with a speed of 1.7 m s^{-1} at 30° to the original line of motion, while the second disc leaves with a certain speed at a certain angle below original line of motion. Determine the speed of the second disc after the collision.

[5 *marks*]

4.

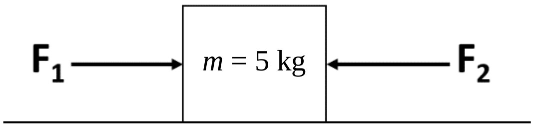

FIGURE 4.1

(a) **FIGURE 4.1** shows a box that is being pushed by Ali and Azryq on a frictionless horizontal floor. Ali pushes the box with force, F_1 = 10 N to the right while Azryq pushes the box with force, F_2 = 8 N to the left.

(i) Sketch free body diagram.

(ii) Calculate the magnitude and direction of the acceleration of the box?

[6 *marks*]

(b)

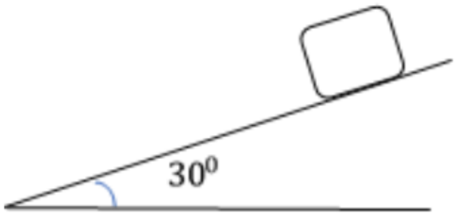

FIGURE 4.2

FIGURE 4.2 shows a 10 kg object slides down on an inclined plane and the coefficient of friction between the inclined plane and the object is 0.1. Calculate

(i) normal force,

(ii) frictional force and

(iii) total horizontal force experienced by the object.

[6 *marks*]

5. (a) A 180 g ball is tied to the end of a cord and rotated in a horizontal circle of radius 1.20 m and experienced a centripetal acceleration of 6.5 m s^{-2}. Assuming that the cord is in horizontal (i.e. the gravity can be neglected).

Determine

(i) the angular velocity of the object in rev/min,

(ii) the tension in the cord.

[5 *marks*]

(b) An ambulance experiences a centripetal acceleration of 15 m s^{-2} while traveling around a curve road with a speed of 16 m s^{-1}. At the same time, a 2000 kg car followed the ambulance from behind along the same curve road at a speed of 8 m s^{-1}.

Calculate

(i) the centripetal acceleration the car experienced,

(ii) the minimum friction between the car and the road to make sure it can travel without skidding.

[4 *marks*]

6. (a) Angular velocity of a rigid body decreases uniformly from 600 rpm to 300 rpm in 6.0 s. Calculate

(i) angular acceleration

(ii) number of rotation during that time interval

(iii) additional time needed for rigid body to stop

[9 *marks*]

(b)

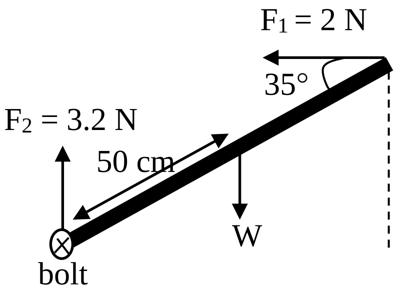

FIGURE 6

FIGURE 6 shows a 100 cm uniform rod with mass 150 g acted by three forces. Calculate total torque of the system.

[3 *marks*]

7. (a) A metal rod has a length of 0.4 m and a cross-sectional area of 4 cm^2. At a steady state, the temperature different is 27 K. Given the thermal conductivity, k of the metal rod is 385 J m^{-1} K^{-1}. Calculate the

(i) temperature gradient along the rod.

(ii) total amount of heat conducted in 2 minutes.

[4 *marks*]

(b) A fresh air is composed of Nitrogen gases, N_2 and Oxygen gases, O_2. Determine the rms speed of nitrogen and oxygen gases at 45 °C.

(Given the molar mass, *M* of nitrogen gas = 0.028 kg mol^{-1} and the molar mass, *M* of the oxygen gas = 0.032 kg mol^{-1}).

[4 *marks*]

(c) A gas contained in a cylinder fitted with a frictionless piston expands against a constant pressure of 1 atm from a volume of 0.03 m^3 to a volume of 0.07 m^3. In doing so it absorbs 400 J of thermal energy from its surroundings. Calculate

(i) the work done by gas.

(ii) the change in internal energy of system.

[4 *marks*]

(d)

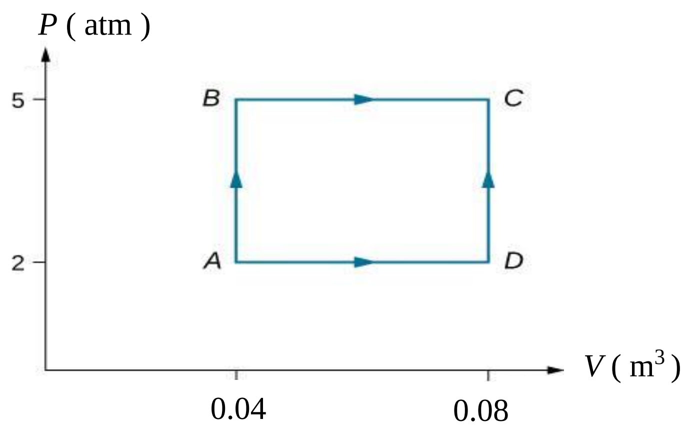

FIGURE 7

Consider the processes shown in **FIGURE 7** below. In the processes AB and BC 3600 J of heat are added to the system, respectively. Calculate

(i) the work done in the processes AB and BC and state either the work is done **on** the system or **by** the system on this process.

(ii) the internal energy change in process BC.

[6 *marks*]

END OF QUESTION PAPER

KERTAS SOALAN TAMAT